spatialHeatmap: Visualizing Spatial Assays in Anatomical Images and Network Graphs

Jianhai Zhang, Jordan Hayes, Le Zhang, Bing Yang, Wolf B. Frommer, Julia Bailey-Serres, and Thomas Girke

Girke Lab, University of California, Riverside

Introduction

The *spatialHeatmap* package provides functionalities for visualizing cell-, tissue- and organ-specific data of biological assays by coloring the corresponding spatial features defined in anatomical images according to a numeric color key. The color scheme used to represent the assay values can be customized by the user. This core functionality of the package is called a spatial heatmap (SHM) plot. It is enhanced with visualization tools for groups of measured items (*e.g.* gene modules) sharing related abundance profiles, including matrix heatmaps combined with hierarchical clustering dendrograms and network representations (Langfelder and Horvath 2008). The environment also allows to co-visualize single cell assay data with various embedding algorithms of high-dimensional data. Automated detection of spatially enriched abundance differences is also supported, such as genes expressed in specific tissues.

The functionalities of *spatialHeatmap* can be used either in a command-driven mode from within R or a graphical user interface (GUI) provided by a Shiny App that is also part of this package (Chang et al. 2021). While the R-based mode provides flexibility to customize and automate analysis routines, the Shiny App includes a variety of convenience features that will appeal to experimentalists and other users less familiar with R. Moreover, the Shiny App can be used on both local computers as well as centralized server-based deployments (*e.g.* cloud-based or custom servers) that can be accessed remotely as a public web service for using *spatialHeatmap*'s functionalities with community and/or private data. The functionalities of the *spatialHeatmap* package are illustrated in Figure 1.

Graphical Overview

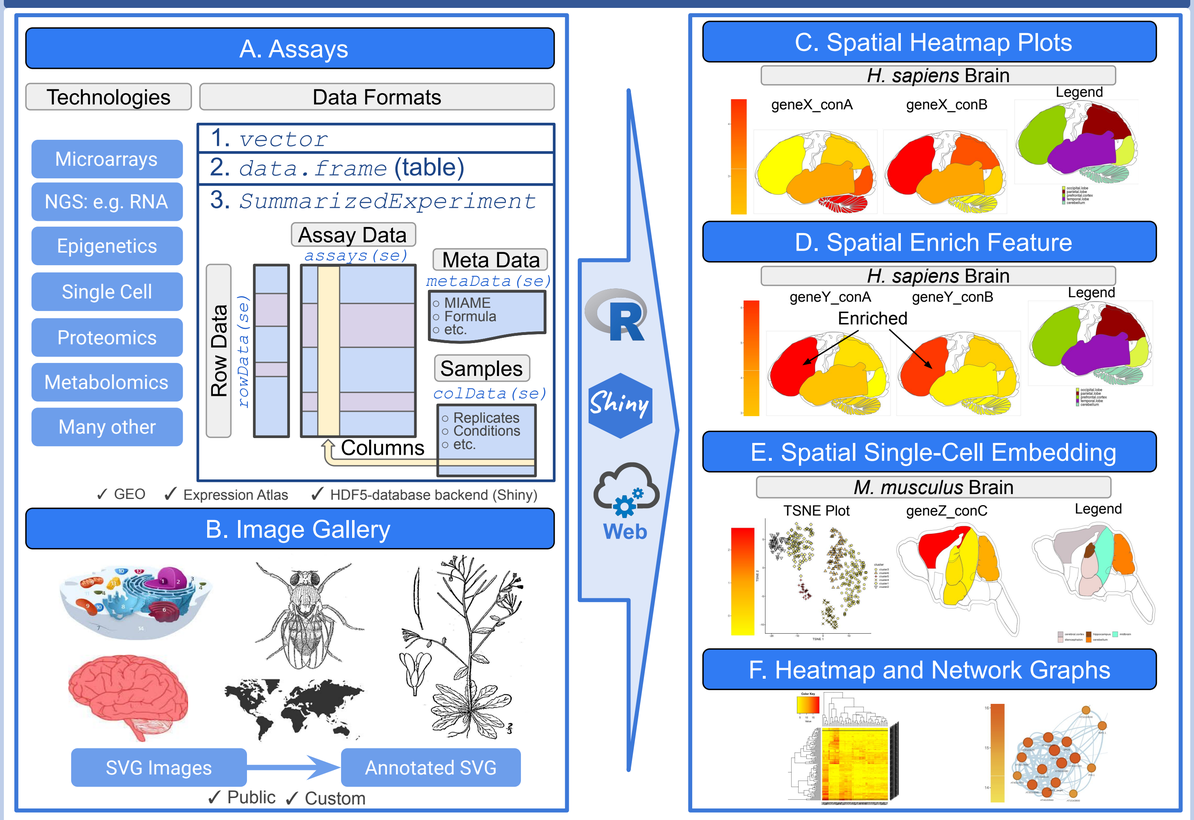

Figure 1: Overview of *spatialHeatmap*. (A) The *saptialHeatmap* package plots numeric assay data onto spatially annotated images. A wide range of omics technologies is supported including genomic, transcriptomic, proteomic and metabolomic profiling data. The assay data can be provided as numeric vectors, tabular data, or *SummarizedExperiment* objects. The latter is a widely used data container for organizing both assay data as well as associated annotation and experimental design data. (B) Anatomical and other spatial images need to be provided as annotated SVG (aSVG) files where the spatial features and the corresponding data components of the assay data have matching labels (e.g. tissue labels). (C) The assay data are used to color the matching spatial features in one or more aSVG images according to a color key. The result is called a spatial heatmap (SHM) or spatiotemporal heatmap (STHM) plot. Multiple measurements can be visualized in the same plot, such as several factors (*e.g.* genes, proteins, metabolites), treatment conditions, growth stages and more. (D) Spatial abundance differences of assayed features (RNAs) can be auto-detected with several algorithms. (E) Single cell data can be co-visualized using several embedding algorithms, such as tSNE and UMAP. (F) Data mining graphics, such as matrix heatmaps and network graphs, are integrated to facilitate the identification of factors with similar assay profiles. The functionalities of can be accessed from local computers via the R console or a graphical user interface based on Shiny. In addition, the latter can be deployed as a web service on custom servers or cloud-based systems.

References

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SVG repository: https://github.com/ebi-gene-expression-group/anatomogram/tree/master/src/svg

Spatial Heatmap Plots

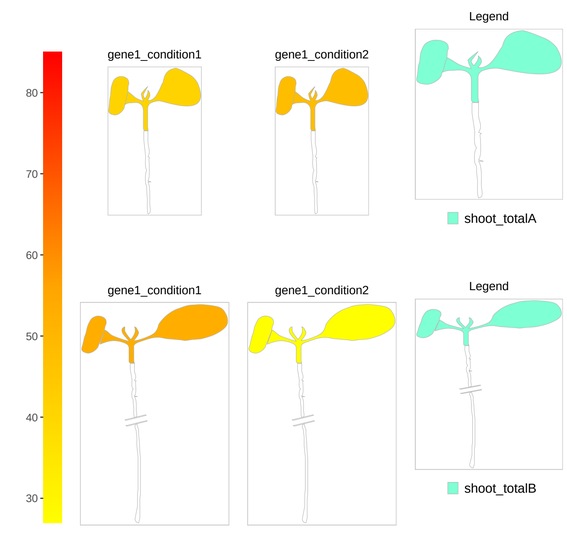

Figure 2: Regular spatial heatmap of *Arabidopsis thaliana* seedlings at two growth stages. The expression profiles of *gene 1* under *condition 1* and *condition 2* are plotted for two growth stages (top and bottom row).

Spatial Enrich Feature

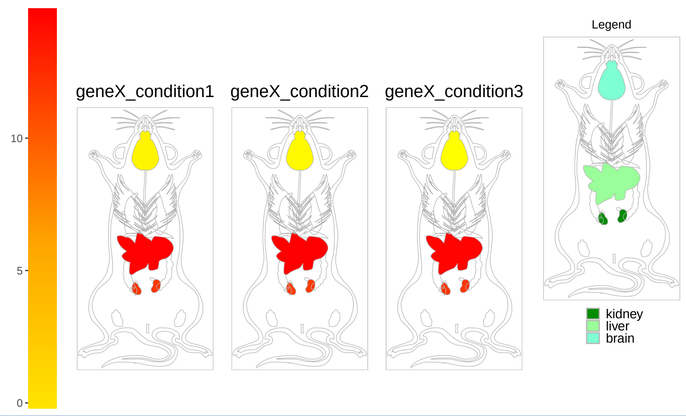

Figure 3: Spatially enriched gene expression signals are plotted onto the corresponding anatomical regions of mouse.

Spatial Single-Cell Embedding

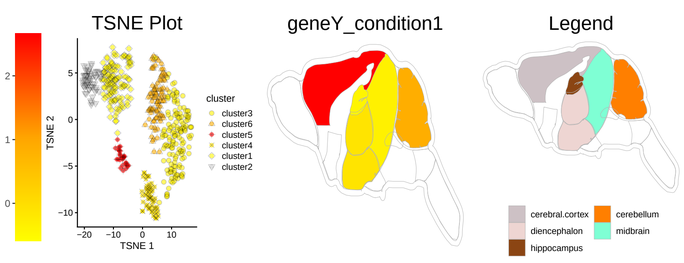

Figure 4: Co-visualization of single cell data in mouse brain. Single cell clusters are presented in form of a tSNE plot. The cell types are associated via with the corresponding spatial features in the corresponding spatial heatmap plot via a common color scheme.

List of Important Functionalities

Function Name	Description
spatial_hm	Main function. Maps data to matching spatial features to generate SHMs. Provide flexible usage such as sets selected tissues transparent, organizes SHMs by genes or condition, takes customized color scheme, etc.
spatial_enrich	Compares target tissue with every other reference tissue to detect tissue-specific signals.
cluster_cells	Clusters cells to define cell groups, whose expression profiles could be mapped to SVG images.
deconvolve	Given a scRNA–seq and bulk RNA–seq data, the latter is decomposed into different cell types by using the former as reference.
return_feature	Query and download aSVG files from online repos.
update_feature	Update feature identifiers to match with data.
matrix_hm	Visualize target genes in the context of nearest neighbors that sharing most similar expression profiles in form of static and interactive matrix heatmaps.
network	Visualize target genes in the context of nearest neighbors that sharing most similar expression profiles in form of static and interactive networks.
adj_mod	Identify network modules, which are clusters of genes showing highly similar expression profiles.
shiny_all	Launch the interactive Shiny app, where all functionalities are integrated on the same page. Would appeal to non-R users.
custom_shiny	Create customized Shiny app by including custom data/aSVGs, a config file.
write_hdf5	Store a batch of data sets in HDF5 databases, which can serve as database backend in custom Shiny app.
norm_data	Convenience function for normalizing sequencing data.
aggr_data	Convenience function for aggregating "sample/conditon" replicates in data.
filter_data	Convenience function for filtering out unreliable data.

Table 1: Descriptions of selected R functions.

Future Work

Multi–omics application: Integrate data of proteomics and metabolomics on top of genomics.

Availability

Bioconductor: `https://bioconductor.org/packages/devel/bioc/vignettes/spatialHeatmap/inst/doc/spatialHeatmap.html`
Github: `https://github.com/jianhaizhang/spatialHeatmap`
Shiny app: `https://tgirke.shinyapps.io/spatialHeatmap/`

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